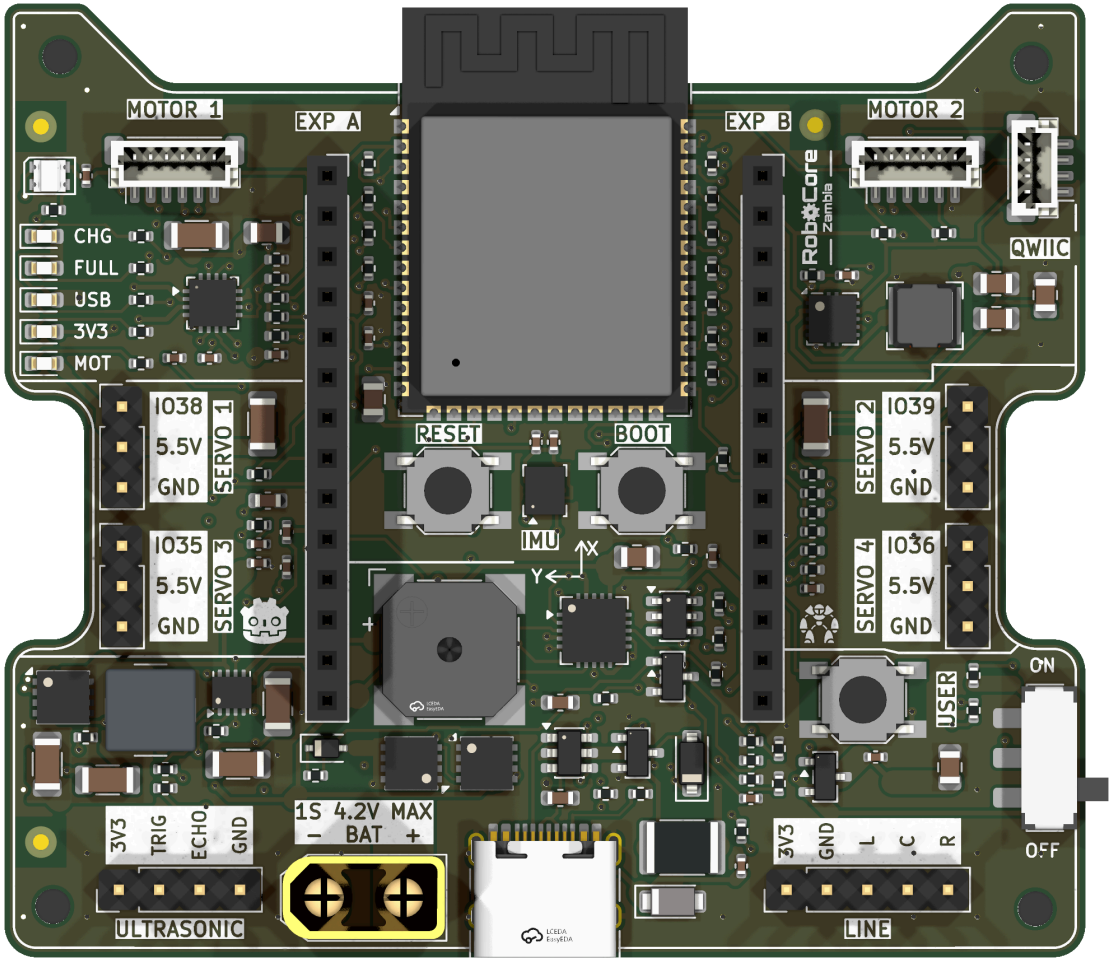
RoboCore V2 · robot controller, rev 2.0



AT A GLANCE

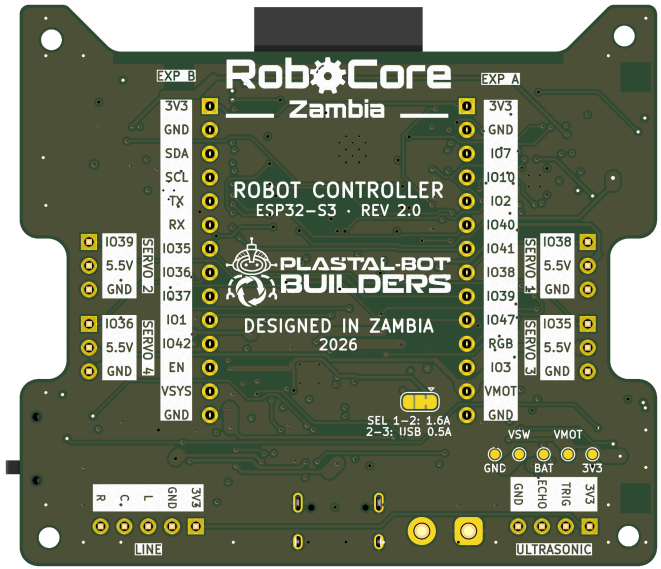
Board	68 × 60 × 1.6 mm, 4 layers, ENIG, green mask / white silk
Brain	ESP32-S3 (dual-core 240 MHz, 8 MB flash, 2 MB PSRAM)
Drives	2 DC motors with encoders, 4 servos, buzzer, RGB LED
Senses	IMU, 3-channel line follower, ultrasonic, battery voltage, motor current
Power	1S Li-ion pack (XT30), USB-C charging, 3.3 V + 5.5 V rails
Mounting	4 × M2 holes (2.2 mm), side notches for the servo cables
Design	Open-STEM XRP-compatible pinout, drawn from scratch in KiCad 10

MAIN PARTS

Function	Part	Note
MCU	ESP32-S3-WROOM-1-N8R2	Wi-Fi + BLE, native USB (one-click upload). Alternates N16R2, N8; never R8/R16V (octal PSRAM takes IO35–37)
Motor driver	TI DRV8411A (WQFN-16)	Both motors, 4 A peak; current limit 1.0 A (JST-SH contact rating)
Motor connectors	JST SH 6-pin ×2	XRP motor plug: M+, M–, 3.3 V, Enc A, Enc B, GND
IMU	ST LSM6DSOX (LGA-14)	6-axis; mounted at the board centre, over the wheel axle. Alternate LSM6DSO32
3.3 V rail	TI TPS63001 buck-boost	Up to ~1.2 A from a 1S cell, holds 3.3 V down to a flat cell
Motor / servo rail	TI TPS61089 boost → 5.5 V	SG90-type servos stay inside 4.8–6 V; Q102 P-FET cuts the boost when OFF
Charger	MCP73871 (USB-C)	1 A charge, power-path; charging works with the switch off. JP101 selects 1.65 A / 500 mA
Battery input	XT30 + reverse/over-voltage cut-off	1S Li-ion only (3.0–4.2 V). Q103/Q104 + TPS3700 survive a reversed or 2S/3S pack
Power switch	C&K JS102011 slide switch	Right edge; grounds the P-FET gates. OFF draws ~0.03 mA
USB-C	HRO TYPE-C-31-M-12 + ESD	Charging and programming
Lights	WS2812B-2020-V6 RGB + 5 status LEDs	CHG, FULL, USB, 3V3, MOT
Sound / buttons	CMT-8504 buzzer; RESET, BOOT, USER	PWM tones; USER on IO42
Ports	Qwiic (I2C), 2 × 14-pin expansion	3.3 V, VMOT, switched battery, spare GPIOs, UART, I2C

Top side — all SMD parts are on top; white knockout boxes carry the pin names of every header (servos, ultrasonic, line follower, battery)

Connectors and pinout



Bottom side — artwork, test pads, charge-current jumper JP101 and the pin names of both expansion headers

IR inputs use ADC1, which keeps working with Wi-Fi on. IMU axes: +X towards the top edge, +Y towards the left edge (firmware remaps).

CONNECTORS

Ref	Name	Type	Pins / notes
J301 / J302	Motor 1 / 2	JST SH 6	M+, M–, 3V3, ENC A, ENC B, GND (XRP order)
J401 J407 / J402 J408	Servo 1 3 / 2 4	2.54 mm 1×3 ×4	Signal, 5.5 V, GND — 1 + 3 on the left notch, 2 + 4 on the right
J403	Ultrasonic	2.54 mm 1×4	3V3, TRIG, ECHO, GND (HC-SR04 order: straight cable). 3.3 V module (HC-SR04+, RCWL-1601)
J404	Line follower	2.54 mm 1×5	3V3, GND, L, C, R — 3-channel TCRT5000-type module, analog outputs
J405 / J406	Expansion A / B	1×14 female ×2	See the pin order table
J201	Qwiic	JST SH 4	GND, 3V3, SDA, SCL
J102	Battery	XT30	1S Li-ion only, 4.2 V max; 1S2P/3P/4P packs need no change
J101	USB-C	16-pin	Charging + programming
JP101	Charge current	solder jumper (back)	1-2 adapter 1.65 A (default), 2-3 USB 500 mA
TP101–105	Test pads (back)		3V3, BAT, GND and rails

ESP32-S3 GPIO MAP

GPIO	Signal
IO0	BOOT button
IO1	IMU INT1
IO2	IR right (ADC)
IO3	Expansion (strapping pin)
IO4	Battery voltage (ADC, ÷2)
IO5 / IO6	Motor 1 / 2 current sense (ADC)
IO7	IR left (ADC)
IO8 / IO9	I2C SDA / SCL (IMU, Qwiic)
IO10	IR centre (ADC)
IO11 / IO12	Motor 1 IN1 / IN2
IO13 / IO14	Motor 2 IN1 / IN2
IO15 / IO16	Encoder 1 A / B

EXPANSION HEADERS (PIN 1 AT THE TOP)

Pin	J405 EXPANSION A	J406 EXPANSION B
1	3.3 V	3.3 V
2	GND	GND
3	IR left (IO7)	I2C SDA (IO8)
4	IR centre (IO10)	I2C SCL (IO9)
5	IR right (IO2)	UART TX (IO43)
6	Ultrasonic TRIG (IO40)	UART RX (IO44)
7	Ultrasonic ECHO (IO41)	Servo 3 (IO35)
8	Servo 1 (IO38)	Servo 4 (IO36)
9	Servo 2 (IO39)	IO37
10	Buzzer (IO47)	IMU INT1 (IO1)
11	RGB LED data out	USER button (IO42)
12	IO3	EN (reset)
13	VMOT 5.5 V	VSYS_SW (switched battery)
14	GND	GND

GPIO MAP (CONTINUED)

GPIO	Signal
IO17 / IO18	Encoder 2 A / B
IO19 / IO20	USB D– / D+
IO21	Motor driver nFAULT
IO35 / IO36	Servo 3 / 4
IO37	Expansion B
IO38 / IO39	Servo 1 / 2
IO40 / IO41	Ultrasonic TRIG / ECHO
IO42	USER button
IO43 / IO44	UART TX / RX → expansion
IO47	Buzzer (PWM)
IO48	WS2812 RGB LED
IO45 / IO46	strapping pins, unused